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Assessment of black globe thermometers employing various sensors and alternative materials

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ABSTRACT

The black globe temperature (T_g) is employed in thermal comfort studies. It is mainly used to compute the Mean Radiant Temperature (*MRT*) required in bioclimatic analysis. A set of low cost Black Globe Thermometers (*BGT*) was developed to establish a suitable combination of temperature transducers, materials and globe diameters. The dynamic and static responses of the *BGT* were studied. A total of 18 *BGT* was deployed along with a data acquisition system based in the Arduino open-source platform. It was analyzed the influence of the globe material (metallic and polyethylene), globe diameter (100 mm and 67 mm for the metallic and 78 mm for polyethylene) and six types of temperature transducers (K type thermocouple, infrared temperature sensor, thermistor, RTD and two semiconductor-based transducers). It was observed that T_g estimation and response time were transducer and stimuli dependent. The *BGT* presents good responses for indoor measurements and it was established that the most accurate combination was achieved by the DS18B20 digital temperature sensor with a metallic globe of 67 mm of diameter.

1. Introduction

The black globe thermometer (*BGT*) was first introduced by H.M. Vernon in 1932, in an attempt to quantify the overall net effect of thermal radiation over a human body (Vernon, 1932). The instrument adds all radiation sources in a room, both positive and negative. Its readings indicate the combination of air temperature and radiation affecting the human subject (Vernon and Warner, 1932). The instrument proposed by Vernon consisted of a 6 inch (152.4 mm) copper globe, matt black painted with an ordinary thermometer fixed in such a way that its bulb was located at the center of the globe.

The black globe temperature (T_g) is defined as the temperature measured at the center of a thin-walled sphere (globe) with black-plain color; the measurement range is 20–120 °C, employing a non-standarized temperature sensor with an accuracy of ± 0.5 °C between 20 and 50 °C and ± 1 °C over 50 °C (ISO, 1998). A 150 mm globe diameter is suggested although the temperature sensor and globe materials are not defined.

T_g is used to compute the mean radiant temperature (*MRT*) along with air temperature, relative humidity and air velocity; they are considered the basic environmental variables that affect human response to thermal environments (Fanger, 1970). The *MRT* is defined as the uniform temperature of an imaginary black enclosure in which its occupants will exchange the same amount of radiant heat as the real non-uniform enclosure (ASHRAE, 2017; ISO, 1998). This variable is used to determine a large number of comfort, heat/cold stress and bioclimatic indices and models such as Predicted Mean Vote index (PMV) (d'Ambrosio Alfano et al., 2017; Fanger, 1970; ISO, 2005), Standard Effective Temperature (SET) (ASHRAE, 2004), Wet–Bulb Globe Temperature (WBGT) (ACGIH - American Conference of Governmental Industrial Hygienists, 2011; d'Ambrosio Alfano et al., 2014; ISO, 2017), Predicted Heat Strain model (PHS) (d'Ambrosio Alfano et al., 2016; ISO, 2004; Malchaire et al., 2001), Insulation Required index (IREQ) (D'Ambrosio Alfano et al., 2013; Holmer, 1984; ISO, 2007), Physiological Equivalent Temperature (PET) (Höppe, 1999; Mayer and Höppe, 1987) and Universal Thermal Climate Index (UTCI) (Bröde et al., 2012).

For its direct measurement pyranometers and pyrgeometers are employed to quantify short and long wave infrared radiation. However, *BGT* is the instrument most commonly used due to its simplicity, compact equipment, easy calculation, and low cost in comparison with other methods (d'Ambrosio Alfano et al., 2013; ASHRAE, 2017; ISO, 1998; Parsons, 2006). Also, some equations had been proposed that

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Table 1
Characteristics of the temperature sensors employed inside the black globe thermometers.

Sensor	Reference	Physical principle	Accuracy	Justification
Thermocouple (*TC*)	Type K, MAX6675	Thermoelectric Seebeck effect	± 2°C	Most employed sensor in the cited literature.
Digital and infrared (*IR*)	MLX90614	Thermopile Seebeck effect	± 0.5°C	Good noise rejection compared with analog sensors
Digital thermometer (*DS*)	DS18B20	Integrated circuit based on silicon	± 0.5°C	Good noise rejection compared with analog sensors
Integrated circuit (*LM*)	LM335A	Integrated circuit based on silicon	± 0.5°C	Most employed sensor in the cited literature
10 kΩ thermistor (*TM*)	MF52	Semiconductor (oxide) electrical resistance change	± 0.2°C	Most employed sensor in the cited literature
RTD PT100 (*PT*)	PT100, HX711	Metal electric resistance change	± 0.5°C	The same sensor employed in the HD32.2 *BGT*

allow to estimate with precision T_g in terms of environmental variables generally measured (Dimiceli et al., 2013; Turco et al., 2008).

The T_g assumed at equilibrium results from a balance between heat gained and lost by radiation and convection (ASHRAE, 2017). Through an energy balance on the globe thermometer a relation between T_g and *MRT* has been established (Fountain, 1987).

Studies have been conducted to determine the impact of air velocity, size and material of the globe in the measure of T_g. Bedford and Warner (1934) proposed an expression to estimate *MRT* as function of air temperature and air velocity. Hey (1968) studied the relation between globe diameter and air velocity finding that the sensitivity of small instruments to radiation was not seriously reduced and could be readily estimated with small instruments once air velocity was known.

Some other studies suggest that globes with alternative diameters and materials can be used to compute *MRT*. Humphreys (1977) suggested that a globe with 40 mm diameter was more convenient and faster than one with 6 inch (152.4 mm) diameter for low air velocities. While McIntyre (1980) concluded that it was suitable to employ globes with larger diameter when air velocity increases. These results were also confirmed by Quintela et al. (2014).

Fountain (1987) compared the response of six different globes employing static air conditions. It was concluded that the 38 mm table tennis ball had the best time response. For this same diameter, d'Ambrosio Alfano et al. (2014) developed an equation to compute T_g and the WBGT index in function of air velocity, diameter and air temperature. The results were verified experimentally with good agreement.

Alternative globes, like acrylic table-tennis balls have been employed as an economical alternative to implement portable *BGT*s (da Silveira Hirashima and Sad de Assis, 2011), due to their mechanical characteristics. These balls presented a shorter response time than a metal *BGT* with a 150 mm globe diameter (Fountain, 1987; Nikolopoulou et al., 1999). But in environments with heterogeneous radiation, such as outdoors, there were significant differences between the temperatures measured with metal and acrylic globes, which leads to errors in *MRT* computation. The temperature inside a non-metallic globe was not distributed homogeneously, therefore the temperature in the center did not correspond to the average of the temperature of its surface. This phenomenon did not occur with metallic materials due to its high thermal conductivity. Another cause of this was attributed to a selective energy absorption at short wavelengths by the non-metallic material (Nikolopoulou et al., 1999; Wang and Li, 2015). Hence, it was recommended to avoid the use of materials with low thermal conductivity and to use metal spheres with wall thicknesses less than 0.5 mm to reduce the response time (Fontana, 2010).

Temperature transducers with electric responses are being employed to measure the temperature inside the globes. The most implemented sensors referred in literature are thermocouples (Fontana, 2010; Fountain, 1987; Liang et al., 2013; Nikolopoulou et al., 1999; Wang and Li, 2015), followed by thermistors (Purswell and Davis, 2008; Zhang et al., 2016) and semiconductors, like the LM335 (Hirashima, 2010). It is necessary to compare these sensors performance measuring T_g. It is also lacking to establish a selection criterion between the different low-cost sensors and how the dynamic response of these transducers affects the measurement of T_g.

In this work six low cost temperature transducers were evaluated in order to determine its suitability for T_g measurement. Employing different globe diameters and materials,the correlations between transducer, material and diameter were established. *BGT* time response, stability, and accuracy were studied at different excitation levels at room temperature, in a closed greenhouse and employing infrared lamps. To validate the measurements, a commercial thermal stress meter equipment, according to ISO (1998), was employed. The data acquisition system for these experiments tend to be expensive (Ahmad et al., 2016). An Arduino open-source electronic platform was employed as a data acquisition system, equipped with a microSD saving capabilities, real time clock and Bluetooth communication, bringing flexibility to the sensor network.

2. Materials and methods

To implement the black globe thermometers, six different types of temperature sensors were used (Table 1). Those sensors were installed in the center of three different types of globes (Table 2), resulting in 18 *BGT*s (BGT_c). The exterior of the globes was painted with matte black spray commercial paint to increase the absorptivity. The interior of the globes used with the infrared sensors was also painted, in order to not affect its calibration. As a reference sensor a *BGT* DELTAOHM HD32.2 thermal stress meter (BGT_p) (Fig. 1) was used. It has a *BGT* (TP3275) with 150 mm globe diameter and a PT100 RTD temperature transducer, a dry-bulb thermometer (TP3207) and a natural wet-bulb thermometer (HP3201). All the sensors have ± 0.15 °C of accuracy. This sensor was calibrated in an ISO certified laboratory and its data was saved in a datalogger with 15 s of sampling rate. The testing *BGT*s were randomly distributed on a horizontal platform, at a distance of 40 cm above the horizontal platform, and each sensor was spaced 30 cm from each other (Fig. 1 - Top right). The platform was placed 30 cm above the floor.

The accuracy of the three most used transducers, refered in the aforementioned literature, were analyzed. Additionally, the *MLX90614* digital IR thermometer (IR_O) was employed. This instrument has a temperature body sensor (IR_A) to compensate the IR measurement. The IR sensors, that has 80° of field of view, were located with the thermopile upwards. They were employed to establish the temperature gradient at the interior of the globes, allowing to compare the behavior of the different globe materials.

A low-cost data acquisition system, based on the open-source Arduino Mega 2560 development platform, was built to acquire the signals from the sensors. In addition, a *SPI* module for microSD memory card was used for data storage. A *HC05* Bluetooth module and a *DS1307* real-time clock module were also implemented to monitor the operation of the sensors remotely and to keep track of the real-time data

Table 2
Characteristics of the globes used in the black globe thermometers.

Material		Diameter	Thickness
Stainless steel	(*BS*)	100 mm	0.7 mm
Stainless steel	(*SS*)	67 mm	0.7 mm
Polyethylene	(*PS*)	78 mm	0.2 mm

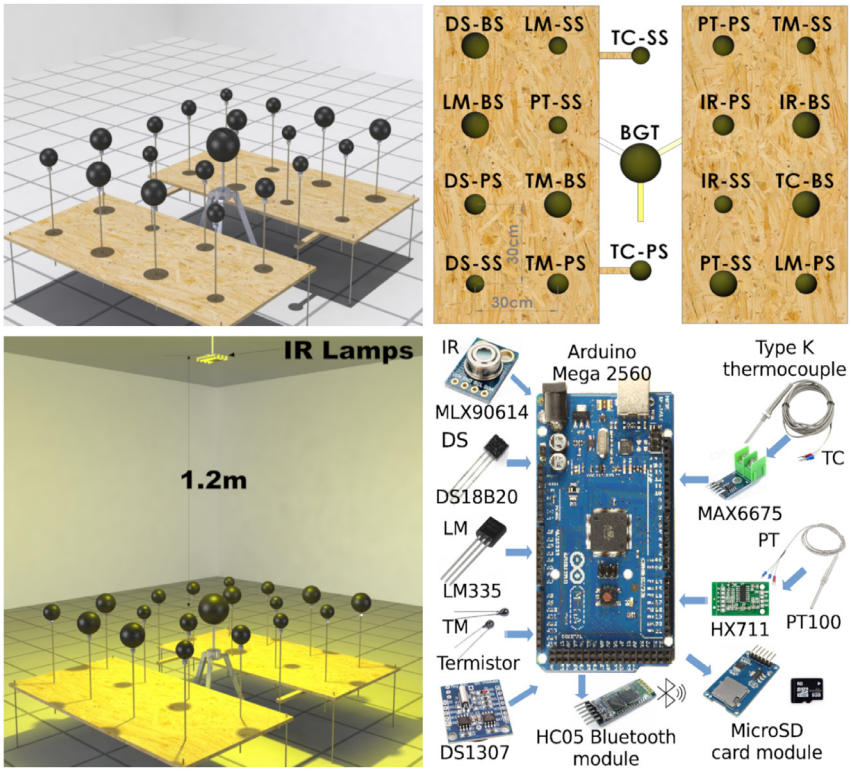

Fig. 1. Top left - Experimental layout of the black globe thermometers developed. Top right - Spatial distribution of black globe thermometers (BGT_c) and the reference thermal stress measurement device (HD32.2 Delta Ohm) used as patron (BGT_p). Bottom left - Experimental layout of IR lamps test employing 4 infrared lamps of 200 W each as a thermal radiation source located 1.2m above the *BGT* s. Bottom right - Layout of the data acquisition system developed based on Arduino electronic-platform.

acquisition. In Fig. 1 (Bottom right) a scheme of the developed system with all its components is presented.

Through Arduino's development environment (Arduino.cc, 2018) and using its own programming language, a real-time data acquisition algorithm was programmed. The libraries *SPI* and *SD* were used to establish the communication with the microSD memory module. The library *Wire* was used to implement the *I2C* communication protocol required by the *DS1307* real-time clock module and by the *DS18B20* digital temperature sensors. Open-source libraries available on the Web such as *MAX6675* (Adafruit Industries, 2018) and *RTClib* (Wippler, 2016) were also employed. The first one allows the communication with the *MAX6675* module, used to amplify and condition the analog signal of the thermocouples, whereas the second one was employed to configure the *DS1307* real-time clock.

Other libraries were also implemented, such as *Adafruit MLX90614* (Industries, 2018) used to establish communication with the *MLX90614* IR temperature sensors which measure contactless temperature of an object (T_{IR_O}) and the ambient temperature (T_{IR_A}) around the sensor. The *HX711* library (Bogde, 2016) was employed to calibrate and take advantage of the *HX711* 24-bit analog-digital converter who receives the *PT100 RTDs* analog signals. Finally, the *OneWire* (Studt et al., 2016) and *DallasTemperature* (Burton et al., 2018) libraries were used to establish communication with the *DS18B20* digital temperature sensor through *one-wired* serial protocol.

The temperature measurement was carried out sequentially starting from the thermocouples, then the *MLX90614* infrared sensors, the *DS18B20* digital sensors, the *LM335* analog sensors, the thermistors and finally the *PT100*. The temperature measured by each sensor was computed employing a moving average, obtained from a series of 20 instantaneous measurements. The average temperature value of each sensor was stored in a microSD memory along with the date and time of the acquisition. All the measurements and data logging process took less than 10 s.

A calibration experiment was performed to determine and to correct the deviation of the temperature measurements made by each low cost temperature sensor. The correlation was based on the *PT100* sensor used by the BGT_p. This was necessary even for the factory calibrated *DS18B20* sensors, because smalls deviations in the temperature measurements could hide the effect of the sensor type, globe diameter and material over the estimation of T_g. Additional information with regard to the temperature sensor calibration can be found in the Appendix A.

After the sensors calibration, three experiments were carried out for 60 days to study the relations of the *BGT*s with the ambient temperature variations and the incidence of thermal radiation sources: (1) The *BGT*s were exposed to air temperature variations in a no-ventilated room (Fig. 1 - Top left). (2) The *BGT*s were exposed to a source of thermal radiation composed by 4 infrared lamps of 200 *W* each, located 1.2 m above the *BGT*s in a no-ventilated room (Fig. 1 - Bottom left); the lamps were arranged very close to each other in order to create a single source of thermal radiation. (3) *BGT*s were located at the center of a chapel-greenhouse facility with an area of 16 × 8 m, average height 4 m, built in structural steel and with a plastic cover with HALS stabilizer, 2% of transmission UV coefficient between 290 and 340 nm, 55% thermicity, 85% of total light transmission and 55% of total light diffusion. In all the tests a 15 s sampling rate was used to collect the data.

Due to the setup size it was not possible to find a climatic chamber to carry on the test. A brick empty and closed room was employed. This work is based on the premises that the black globe temperature, measured with the BGT_p (T_{g_p}) at each point where the BGTc's were located, was homogeneous (± 0.5°C of difference among them). Although the measured radiation at each point was not homogeneous, the magnitude of thermal radiation coming from above the globes was bigger that the surrounding, thus the effect of the horizontal obstruction of the globes and the air velocity were considered negligible over T_g.

T_g's residuals computed as the subtraction between BGT_p and BGT_c measurements were analyzed using the Tukey's HSD test. Differences between the BGT_c measurements with regard to the globe material, diameter, sensor type and response time were studied with this test. In contrast to ANOVA or Kruskal-Wallis test, this statistical procedure allows to perform multiple comparisons simultaneously between all the BGT_c pair combinations. It also makes possible to know which BGT_c provided a most accurate T_g measurement. For this test, the open-source

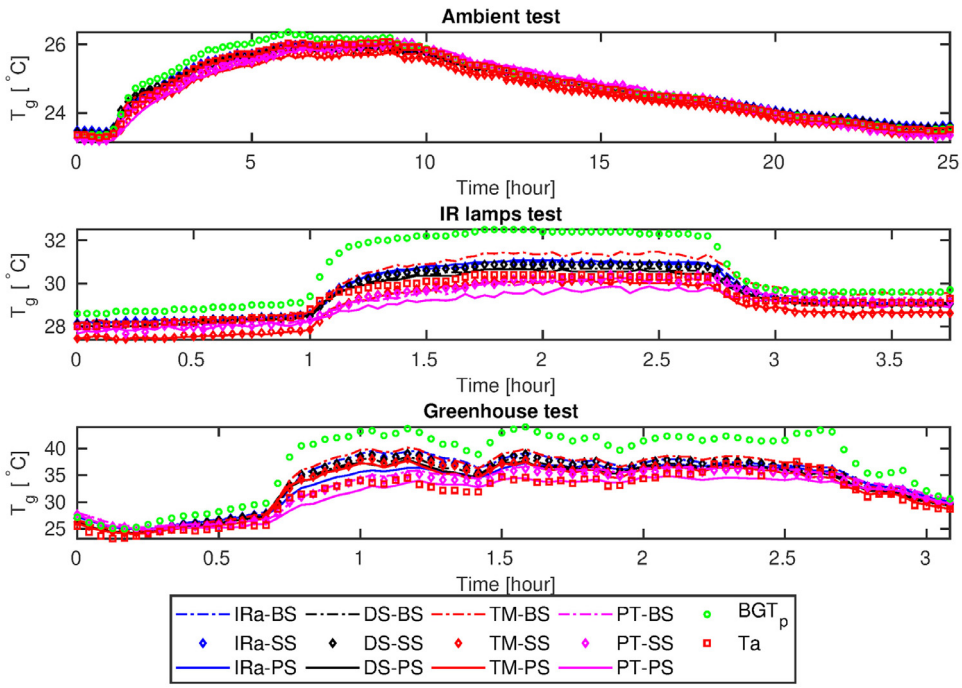

Fig. 2. Zoomed portions of the globe temperatures (T_g) measured by the reference (BGT_p) and developed *BGT*s. Top Ambient test. Middle IR lamps test. Bottom-Greenhouse test.

statistical computational package *R* was used with a confidence level of 0.95.

3. Results and discussion

According to the results of the temperature sensors calibration process (found in the supplementary material), the *TC* and *LM* sensors have a measurement error bigger than 0.5°C. The *TC* signal was extremely noisy while the quantization error of the *LM* was as big as its accuracy. Therefore these two sensors will be discarded in the future analyses. In order to improve the performance of these sensors, it is necessary to consider to use a signal conditioner with more resolution.

When *BGT*s were exposed to different levels of thermal radiation, the BGT_c appropriately follows the dynamic behavior of BGT_p (Fig. 2). However, it is shown that the black globe temperatures measured with the BGT_c (T_{g_c}) were below T_{g_p} for almost all tests.

For the ambient test (Fig. 2 - Top), the differences between T_{g_c} and T_{g_p} increase when the ambient temperature were higher than 28°C, but the T_{g_c} were located around the T_{g_p} values when the ambient temperature continuously decreased. The differences between T_{g_p} and T_{g_c} also depend on the magnitude of thermal radiation, as was evidenced in the IR lamp test (Fig. 3 - Middle). When the IR lamps were on (sharp increase of T_g) these differences were higher than when they were off (sharp decrease of T_g).

In the greenhouse test (Fig. 3 - Bottom) it was evident that the T_{g_c} values were located close to the T_{g_p} when the ambient temperature continuously decreased. At higher ambient temperatures, the differences between T_{g_p} and T_{g_c} increased. This behavior was previously observed by Nikolopoulou et al. (1999) and Wang and Li (2015), which studied the evolution of T_g for *BGT* using copper and plastic globes. Nikolopoulou et al. (1999) argued for selective absorption of wavelengths of thermal radiation as a function of the paint layer applied to the globe and the material used. Whereas Wang and Li (2015) explained this difference in the inability of non-metallic materials to evenly distribute thermal radiation throughout the globe, which generates temperature gradients into the interior of the globes and consequently the inability of the temperature sensor to reproduce appropriately T_g.

When T_g residuals (mathematical difference between T_{g_p} and T_{g_c}) were plotted (Fig. 3), it was evident that the T_g residuals were positives for almost all BGT_c, corresponding to T_{g_c} values lower than T_{g_p}. The BGT_c underestimated T_g. Variations in its measurements depend on the incident thermal radiation and on the type of sensor used, as was stated before. In addition, some BGT_c overestimated T_g up to 0.8°C when the ambient temperature continuously decreases and below 28 °C.

The box diagrams of the T_g residuals shown in Fig. 4 allows to visualize the variability of T_g measurements for each BGT_c. The top whisker of the box plots changes considerably between the tests conducted. On average the residuals increased 70.83 ± 50.82% from ambient to IR lamps test, 267.28 ± 90.32% from IR lamps to greenhouse test and 495.04 ± 101.68% from ambient to greenhouse test. The bottom whisker of the box plots remains around zero and does not change considerably between tests. The BGT_c with *PT* sensors, followed by the *TM* sensors, showed the highest variability, meanwhile the BGT_c with IR_A and *DS* sensors presented a similar behavior on each test with a lower variability.

The IR lamps test allows to compare the behavior of the BGT_c excited with two levels of thermal radiation (IR lamps off and on, Fig. 5) once they reached a steady state. It is shown again that the higher variability for both levels corresponds to the BGT_c with *PT* sensors, followed by the *TM* ones. Meanwhile, the variability of the BGT_c with IR_A and *DS* sensors were considerably reduced. It is also shown, how the BGT_c with *TM* sensor and *BS* globe, despite of its high variability, gave the lower underestimation of T_{g_p}. This last observation could be misunderstood as an effect of the location of the BGT_c with respect to the others, considering that it can receive a higher amount of thermal radiation. However this behavior was also observed in the greenhouse test, where the radiation from above was received equally for all the *BGT*s, therefore, this behavior is related principally to the type of sensor in combination with the type of globe used. On the other hand, the reduction in the variability of the measurements showed in the Fig. 5 in comparison with the Fig. 4 - Middle, could be due to the fact that the transition measurements from lamps OFF to lamps ON and vice versa were not considered. These measurements can provide higher differences between the T_{g_p} and T_{g_c} due to different dynamics of the sensors.

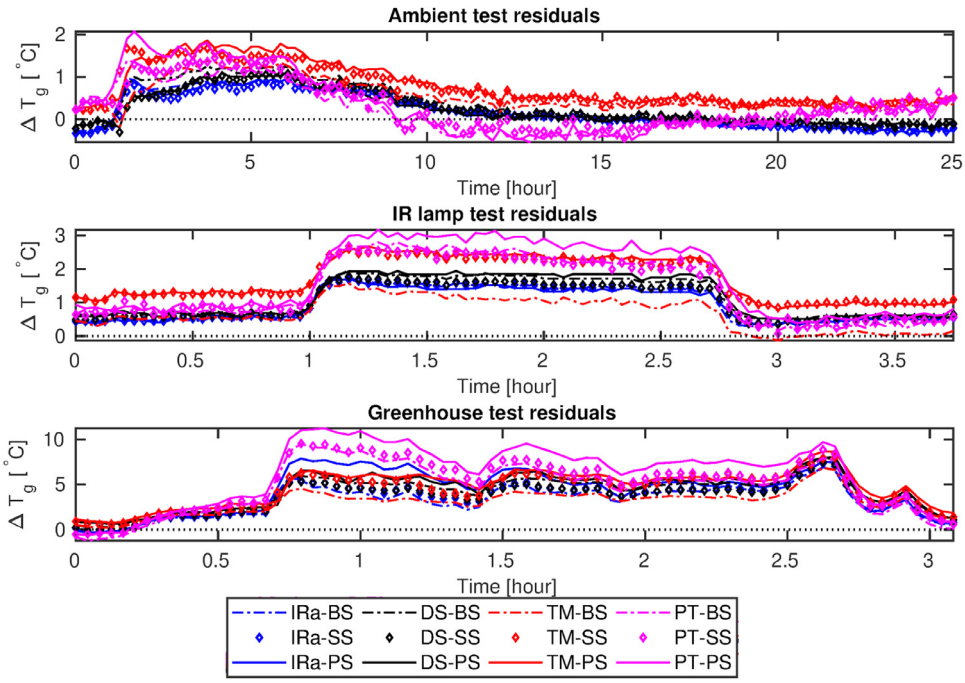

Fig. 3. Globe temperature residuals (ΔT_g) calculated as the difference between reference T_g and T_g for each developed BGT_c. Top Ambient test. Middle IR lamps test. Bottom- Greenhouse test.

3.1. Behavior of globe materials

With respect to globe materials, the behaviors of the BGT_c were analyzed through the maximum difference between T_{g_p} and T_{g_c} which is represented by the top whisker of the box plots for each BGT_c, as shown in Fig. 4. For the BGT_c with IR_Asensors, the estimation of T_{g_p} and its variability were very similar for all the globes in the ambient and IR lamps tests, meanwhile the *BS* and *SS* globes also presented a similar variability in the greenhouse test, where the *PS* globe showed the higher variability. For the BGT_c with *DS* sensors, the *BS* and *PS* globes estimated T_{g_p} similarly, meanwhile the *SS* globe gave the lower variability and a slightly better estimation of T_{g_p}. The behaviors of the *SS* and *PS* globes for the BGT_c with *TM* sensors were very similar for all the tests, but the *BS* globe exhibited the lower variability and a better estimation of T_{g_p} for all the tests. For the BGT_c with *PT* sensors, the *BS* and *SS* globes had similar behaviors for all tests, whereas the *PS* globe presented the higher variability.

The RMSE values (Table 3) give an indication of the precision of the T_{g_p} estimations made by the BGT_c. For all tests, the *PS* globes gave the maximum RMSE value (worst performance) for 10 BGT_c (12 BGT_c in total for each globe), especially in the ambient test in which it was obtained for 3 types of sensors. The *BS* globes gave the maximum RMSE

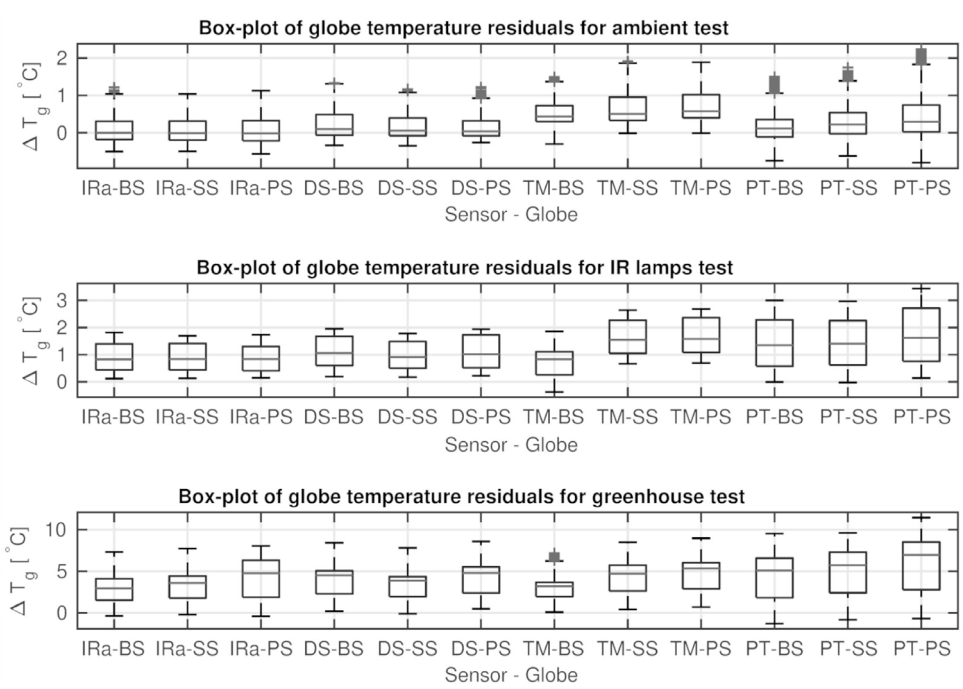

Fig. 4. Box plots of globe temperature residuals (ΔT_g) for each developed BGT_c. Top - Ambient test. Middle - IR lamps test. Bottom - Greenhouse test.

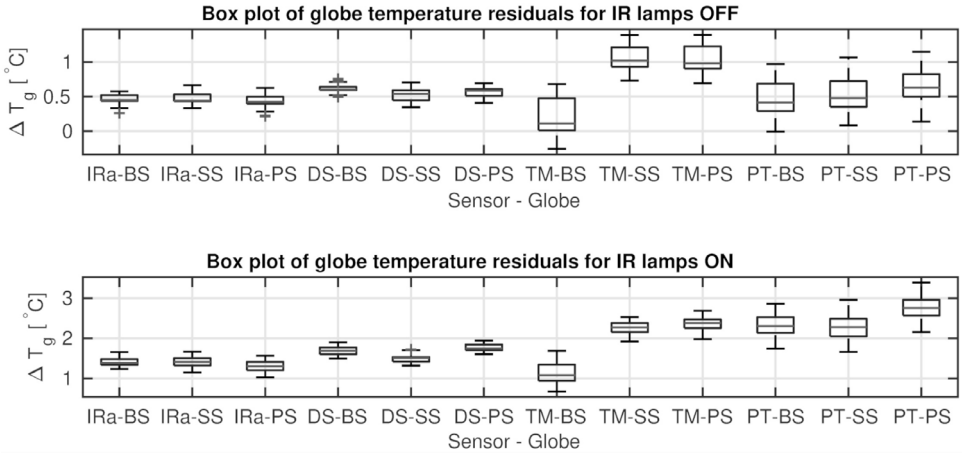

Fig. 5. Box plots of T_g residuals for IR lamp test. Top - IR lamps off. Bottom - IR lamps on.

Table 3
RMSE values in °C for the estimation of T_{g_p} for each BGT_c.

	Ambient test			IR lamps test			Greenhouse test		
BGT_c	***BS***	***SS***	***PS***	***BS***	***SS***	***PS***	***BS***	***SS***	***PS***
IR_A	0.37	0.36	0.40	1.05	1.04	0.98	3.25	3.55	4.77
DS	0.46	0.39	0.38	1.25	1.11	1.28	4.17	3.62	4.49
TM	0.63	0.80	0.84	0.91	1.76	1.82	3.19	4.54	4.99
PT	0.47	0.55	0.71	1.69	1.68	2.01	5.20	5.70	6.74

value for 2 BGT_c; meanwhile for the *SS* globes, no maximum RMSE value was registered.

In the case of the minimum RMSE value (best performance), the *BS* globes gave the minimum RMSE value for six BGT_c, especially for the IR lamps test where three BGT_c were registered; meanwhile, the *SS* globe gave also the minimum RMSE value for six BGT_c, mainly in the IR lamps test where three BGT_c were registered. In contrast, no minimum RMSE value was registered for the *PS* globe. However, a significant difference (p-value = 0.829) between *PS* and *SS* globes for the *DS* sensor in the ambient test after run a Tukey's HSD (honestly significant difference) test to the residuals of BGT_c (differences between the T_{g_p} and T_{g_c} for each sensors) was not found.

Furthermore, the differences between the RMSE values were small in the ambient test and IR lamps test for all the globes and the IR_A and *DS* sensors, and below 1.2 °C in the greenhouse test. The *SS* globe gave the best performance for all tests, providing a precision of 0.4, 1.3 and 3.6 °C for the ambient, IR lamps and greenhouse test respectively.

The high variability of the measurements of the BGT_c with *TM* and *PT* globes led to higher RMSE values. The differences between the RMSE values of all globes were increased, especially in the greenhouse test. For these sensors, higher precisions (lower RMSE values) were obtained when *BS* globes were used, reaching a precision up to 0.6, 0.9 and 3.2 °C for the *TM* sensors and 0.5, 1.7 and 5.2 °C for the *PT* sensors and for the ambient, IR lamps and greenhouse test respectively.

Since in some cases the differences between the RMSE values were small, a Tukey's HSD test ($\alpha = 0.01$) was performed to establish if differences between the BGT_c estimations existed. For the ambient test, a significant difference between the *BS* and *PS* globes of the BGT_c with IR_A sensor (p-value = 0.98) was not found. This result were consistent with the small difference between theirs RMSE values. For the IR lamps test, it was not found a significant difference between all the globes of the BGT_c with IR_A sensor (BS-SS p-value = 1; BS-PS p-value = 0.237 and SS-PS p-value = 0.521); between the *BS* and *PS* globes of the BGT_c with *DS* sensor (p-value = 1); between the *SS* and *PS* globes of the BGT_c with *TM* sensor (p-value = 0.496) and between the *BS* and the *SS* globes of the BGT_c with *PT* sensor (p-value = 1). Finally, for the greenhouse test, significant differences between all the globes of the BGT_c with the same sensors were found. However, a significant difference between the *SS* globes of the IR_A and *DS* sensors (p-value = 0.994) and between the *PS* globes for the same sensors (p-value = 1) was not found.

3.2. Gradient temperature analysis

Through the experimental data obtained in the IR lamps test and the *IR* sensors, it was possible to compare the temperature of the globe's surface and its internal temperature, in order to establish the temperature gradients. The experimental data in steady state were divided in two groups according to the lamps states, IR lamps off and on. The differences between the temperatures registered by IR_O and IR_A sensors were computed. For the IR lamps off state negatives differences for the *BS* and *PS* globes (data not shown) were found, indicating that the temperature at the center of the globe (T_{IR_A}) was slightly above than the surface temperature of the globe (T_{IR_O}), but a less positive difference for the *SS* globe was found. Those differences were below 0.5 °C. Meanwhile, higher positive temperature differences for all the globes were obtained for the IR lamps on state, indicating that T_{IR_O} was above T_{IR_A}. The *SS* globe presented the maximum value (2.7 °C), followed by the *BS* globe (1.7 °C) and the *PS* globe (1.4 °C).

For both IR lamps tests, significant differences (p-value = 0) between T_{IR_A} and T_{IR_O} for each globe were found. The RMSE values presented in the Table 4 indicate the minimum and maximum temperature gradient of the globes, for each state of the for the IR lamps. All the RMSE values for both IR lamps states were significant different (p-value = 0).

Table 4
Average T_{IR_A}, T_{IR_O} and ambient temperature (*Ta*) and RMSE values of the difference between $T_{IR_O}$and T_{IR_A}(values in °C) for each state of the IR lamps

	T_{IR_A}			T_{IR_O}			RMSE(T_{IR_O}-T_{IR_A})			
		Globe			Globe			Globe		
State	***BS***	***SS***	***PS***	***BS***	***SS***	***PS***	***BS***	***SS***	***PS***	*Ta*
OFF	28.78	28.78	28.80	28.37	29.02	28.42	0.4168	0.2502	0.3799	28.71
ON	30.37	30.37	30.48	31.99	33.02	31.77	1.6289	2.6599	1.2916	29.82

The average T_{IR_A} (Table 4) showed similar values for all the globes in each state and significant differences between them (off state *p*-value > 0.99 and on state *p*-value > 0.05) were not registered. However, although T_{IR_A} was close to the ambient temperature, there was a significant difference with it (off state *p*-value < 0.04 and on state *p*-value = 0). The average *SS* globe's temperature surface (T_{IR_O}) was around 0.6 °C above the others globe's surface temperatures for the OFF state and around 1 °C above for the ON states.

The gradients obtained could be explained in terms of the materials used by the globes. The *PS* globe has a material that could work has a thermal isolation, the internal and external surface temperature could be slightly different and the heat is concentrated in an small surface area. On the other hand, the *BS* and *SS* globe are good thermal conductors, the interior and exterior surface temperature was considered equal and the heat was distributed for all the globe surface. However, the *SS* globe reached a higher temperature because it has a small surface area in comparison with the *BS* globe that has a large one.

3.3. Response time

Transient step response changes in thermal radiation levels were obtained from IR lamps and greenhouse tests data. Using the system identification toolbox of MATLAB (The MathWorks, 2013), those step responses were fitted to continuous-time second-order models. As example, in Fig. 6 the step responses of the BGT_p and the BGT_c with *DS* sensor and *SS* globe to changes in the thermal radiation from low (IR lamps off) to high (IR lamps on) level and from high (IR lamps on) to low (IR lamps off) level are shown. The response time (t_r) was considered as the time needed to reach the 95% of total change of T_g (Quintela et al., 2014). The average values of the t_r ($\overline{t_r}$) obtained are summarized in Table 5 for the IR lamps and greenhouse tests respectively.

According to Table 5, for all BGT_c, the $\overline{t_r}$ values were inverse to the thermal radiation received. For the IR lamp test a higher $\overline{t_r}$ than for the greenhouse test was obtained. Particularly, the BGT_p $\overline{t_r}$ average was 17.40 and 8.75 min for the IR lamp and greenhouse tests respectively. The t_r established by the manufacturer was 15 min.

For the IR lamps test, the $\overline{t_r}$ value of all BGT_c were above to 25.64 min, 8.24 min above the $\overline{t_r}$ of the BGT_p, while the BGT_c values with the IR_A sensor and *SS* globe provided the lowest $\overline{t_r}$ (25.64 min). However, significant differences with the *BS* and *PS* globes (*p*-value > 0.99) were not found. The BGT_c with *PT* sensors gave the worst $\overline{t_r}$, particularly with the *BS* globe. This result could be attributed to the high variability of the measurements that were hard to fit appropriate models. In general, for each type sensor tested, the *SS* and *PS* globe gave the best and worst $\overline{t_r}$ respectively.

Table 5
Average response times ($\overline{t_r}$) in minutes for the *BGT*s.Data obtained during several changes in thermal radiation level for the IR lamps and greenhouse tests.

BGT		$\overline{t_r}$	
Sensor	**Globe**	**IR lamps test**	**Greenhouse test**
IR_A	*BS*	26.82	7.92
	SS	25.64	7.75
	SS	27.38	8.30
DS	*SS*	30.65	7.60
	SS	27.35	7.90
	SS	30.84	8.65
TM	*SS*	29.66	8.45
	SS	29.17	7.90
	SS	32.42	8.80
PT	*SS*	38.20	9.05
	SS	37.19	9.25
	SS	38.92	9.50
	BGT_p	17.40	8.75

In greenhouse test, the $\overline{t_r}$ values of the BGT_c with IR_A, *DS* and *TM* sensors were equal or below the BGT_p's $\overline{t_r}$ (8.75 min), whereas for the BGT_c with *PT* the $\overline{t_r}$ were above, for all the globes used. The best $\overline{t_r}$ value was registered by the BGT_c with IR_A sensor and *SS* globe, meanwhile the worst $\overline{t_r}$ value was obtained for the BGT_c with *PT* sensor, in particular for the *PS* globe. Also in this test, for each type sensor tested, the *SS* and *PS* globe gave the best and worst $\overline{t_r}$ respectively.

4. Conclusions

The results of the study have shown that in terms of accuracy and response time, the best Black Globe Thermometer performance was obtained with the internal sensor of the *IR* thermometer and the *DS18B20* digital sensors for both metallic globes. However the use of the DS18B20 sensor is recommended over the internal IR sensor due to its lower acquisition and installation costs. Besides, it is necessary to make offset corrections to the black globe temperature measurement when the black globe thermometer employs globes smaller than the standard (150mm), because the black globe temperature is underestimated, being this underestimation thermal radiation dependent.

The precision in the estimation of the black globe temperature is inverse to the received thermal radiation levels and depends on the type of sensor, material and diameter of the globe. In addition, the response time is also radiation dependent. The temperature differences between the globes increase when the thermal radiation level increases.

The internal temperature of the globes is independent on material and diameter in steady state condition, meanwhile temperature

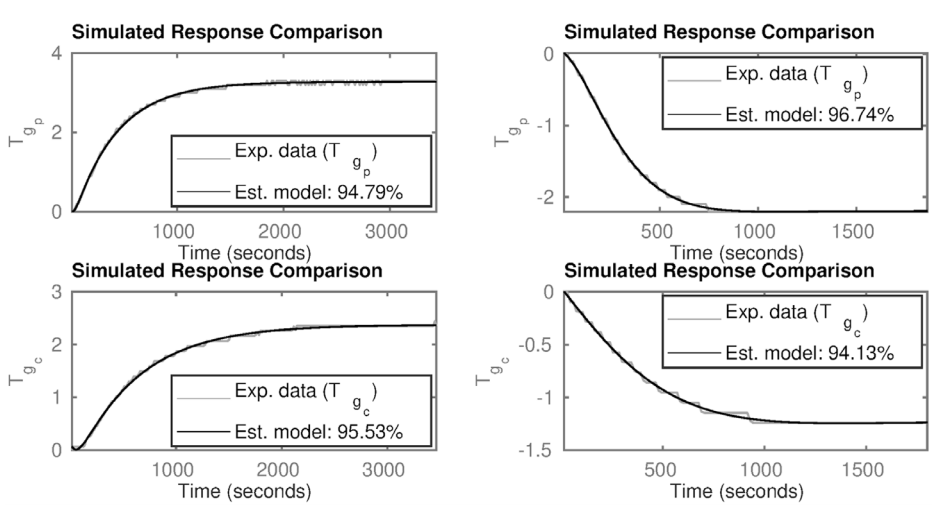

Fig. 6. Second order model system identification fit for a change in the thermal radiation level received. Top - BGT_p. Bottom - BGT_c with *DS* sensor and *BS* globe. Left - Increasing step: lamps OFF to ON. Right - Decreasing step: lamps ON to OFF. The value show in the legend corresponds to the Normalized Root Mean Square Error fitness who represents how close the estimated model output is to the experimental data.

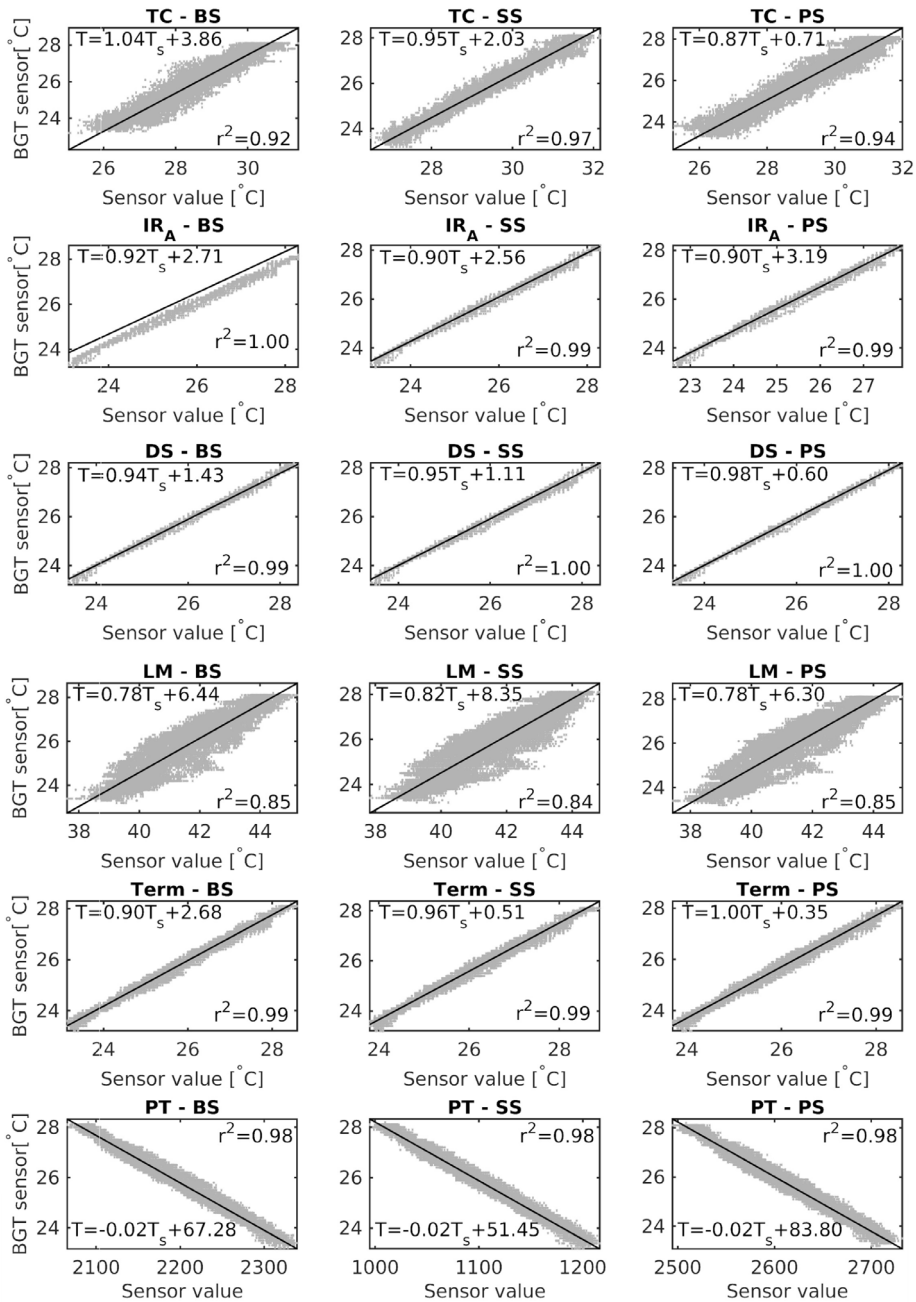

Fig. A.1. Linear regression between the BGT_p temperature naked sensor and each tested temperature sensors. Experimental data (blue dotted) and linear regression models fit (black line)

gradient are dependent of the radiation received.

The acquisition, storage and monitoring system developed presents a good performance, allowing the connection of over 24 sensors in different configurations. This tool can be used in field studies that require the study of temperature gradients or black globe temperature measurements in areas with heterogeneity in air distribution like buildings, greenhouses, sheds or stables.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Temperature sensors calibration

For this experiment, all the sensors were exposed to air temperature by removing the *BGT*'s globes. All sensors were placed at center of a not-ventilated room; only affected by room air ambient temperature changes. This procedure allowed to know the dynamic response of sensors to changes in the room temperature and compare it with BGT_p's *PT100* behavior.

Based on this test, all temperature sensors, except IR_O, were fitted according to the BGT_p's *PT100* temperature sensor measurements by a simple linear regression of the form $T = mT_s + b$, where T is the adjusted temperature (°C), T_s is the measurement provided by the sensor, m is the slope of

Table A.1
Calibrated statistical sensors comparisons. The RMSE values quantifies the variability of the measurements of each temperature sensor. The linear correlation coefficient (r) provides an indication of the linear relationship between the BGT_p temperature sensor and each temperature sensor analyzed.

	RMSE [°C]			r		
BGT_c	*BS*	*SS*	*PS*	*BS*	*SS*	*PS*
TC	0.37	0.22	0.33	0.96	0.98	0.97
IR_A	0.09	0.10	0.12	1.00	1.00	1.00
DS	0.10	0.09	0.07	1.00	1.00	1.00
LM	0.51	0.52	0.51	0.92	0.91	0.92
TM	0.13	0.15	0.15	0.99	0.99	0.99
PT	0.18	0.16	0.18	0.99	0.99	0.99

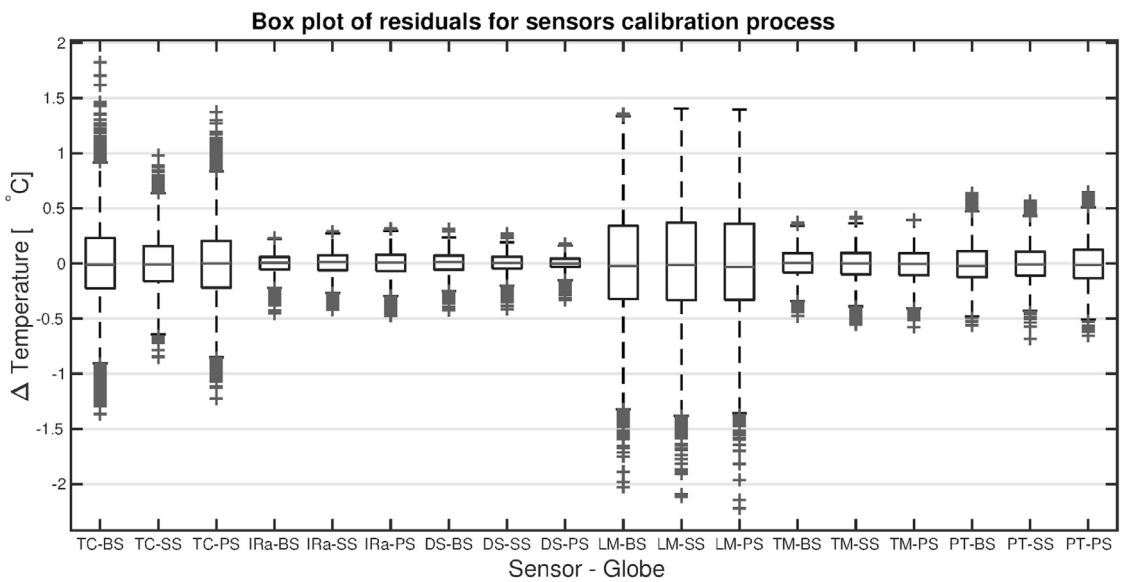

Fig. A.2. Box diagram for the residuals obtained in the sensors fit process. The residuals represent the difference (error) between the temperature provided by the BGT_p naked temperature sensor and the measurements of each temperature sensor analysed.

the linear regression and *b* is the intercept.

In Fig. A.1 a linear correlation between the BGT_p temperature sensor naked and each temperature sensors tested (naked sensors without the globe) is shown. After fitting 70% of the experimental data to linear regression models, the Root Mean Square Error (RMSE) and linear correlation coefficient (r) statistics were computed. The 30% experimental data remaining to compare the goodness of the linear model obtained for each temperature sensor tested (Table A.1). *TC* and *LM* sensors obtain the higher RMSE values and the lower *r* values. There are significant differences compared with the other sensors. Moreover, the box plot of residuals (Fig. A.2) shown that the dispersion of the fitted values of these sensors are higher than 0.5 °C, above the recommendations of ISO (1998). In the case of the *LM* sensor presented a quantization error as big as the sensor accuracy. Whereas for the other sensors the dispersion values fall between ± 0.5 °C, being in accordance with the recommendations of ISO (1998). Given the RMSE, *r* and dispersion range values the *DS* and IR_A sensors present the best behavior.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.agrformet.2019.107891

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